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Use Models and Fidelity Criteria for the Verasonics Arbitrary Waveform Encoders

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Contents

1	Introduction	page 2
2	Who Should Read this Document	page 2
3	Provenance of the Terminology	page 3
4	Why the Encoders Require a Use Model	page 3
5	Types of Use Models	page 4
6	Two Encoder Implementations	page 8
7	Conclusion	page 8
8	References / Bibliography	page 9

1. Introduction

This document explains the **use models** and fidelity metric concepts needed by users of the Verasonics Arbitrary Waveform (AW) Encoder tools. The encoder tools allow the user to specify novel waveforms (subject to bandwidth, power, and duration limits) to be generated by the Verasonics transmitter hardware. The terms “Use Model” and “Fidelity Criterion” are synonymous phrases that determine how the user will define the values of the desired analog waveform. By selecting a Use Model, the user is specifying the domains that influence the reference signal, whether these domains are acoustic or electric, and whether the effects of these domains on the signal are to be mitigated or ignored. The Use Models described here are not exhaustive or unique, but should be useful the large majority of users.

2. Who Should Read this Document

This document is for users who need to transmit a waveform that has a literal description for the reference signal specification as a vector of “analog”, or continuous-valued, RF samples defined at the Verasonics 250MHz transmit clock rate. Note that if the user plans to specify the transmit waveform ONLY as a simple sequence of tri-level states, there is no need to employ this encoder tool, as such a description is directly native to the Verasonics system. One exception to this is when the user already has a tri-level transmit specification appropriate for the intended transducer bandwidth, but desires a rapid change in effective transmitter supply voltage between successive transmit bursts. Here, the term “rapid” means faster than the regulation compliance of

the high-voltage regulator hardware can accommodate. In this case, it is possible to have the encoder generate separate transmit bursts that each approximate the effect of distinct transmit voltage levels. For other transmit waveform specification methods not requiring this encoder, please consult the appropriate section of the Verasonics Sequence Programming Manual.

3. Provenance of the Terminology

The concepts of use models and fidelity metrics are introduced in [Flynn, John A., et al. "Arbitrary waveforms using a tri-state transmit pulser." Ultrasonics Symposium (IUS), 2013 IEEE International. IEEE, 2013.] as follows:

“The fidelity metric employed by the encoder is a design choice that depends on the usage model of the application. [The] NRMSE [normalizer RMS error] metrics considered here measure (1) closeness of the reference waveform to acoustic pressure, and (2) closeness of the reference waveform's predicted acoustic pressure to that achieved. We denote these the one-way DAC synthesis and one-way transducer compensation problems, respectively. Their two-way versions compare the reference to received data, rather than acoustic pressure.”

4. Why the Encoders Require a Use Model

The nature of the encoders is to exploit the filtering inherent in the electrical-to-acoustic conversion process of the transducer. This filtering smooths the 3-level “square-pulse” transmit sequence into an analog acoustic waveform. The encoder is an optimization procedure that determines the digital transmit pulse sequence, given a desired analog waveform to achieve for transmission. Because the Verasonics transmitter is “tri-state” in nature, and so has a physical limitation of three transmit levels, the encoder must optimize the sequence of transmitter states over the transmit interval, in order to achieve the desired analog waveform. An additional “nuisance” parameter, representing system gain, is also estimated.

Note that the desired reference waveform is understood to be achievable only to the extent that its important spectral components lie within the transducer bandwidth.

The term **use model** arises for the following reason. To execute the optimization, the encoder must not only know the reference signal, it must know the location of the reference signal observation point; i.e., where the desired analog or reference “output” signal is to be observed. The encoder must also have a model for the transducer and transmitter dynamics, ultimately in the form of an impulse response (**IR**). The exact form of this IR will thus depend on system components available to influence the reference signal, based on the path taken to the location of its observation point as defined by its use model. The acoustic medium is assumed to be transparent. The various use models are discussed in Section 5.

In Verasonics Sequence Programming scripts, most recognized transducers employ nominal IR data specified by bandwidth and center frequency parameters in the **Trans** object structure. One exception is for the Philips L7-4 transducer, which includes empirically-measured impulse response data in vector form as **Trans** data members.

The **fidelity metric** measure how well the (specified) reference signal “matches” the (predicted) achieved signal due to a hypothetical tri-state sequence, both measured at the observation point. It is a monotonic function of the sum of squared errors. Readers with some exposure to optimization techniques may recognize this as the mathematical notation for **norm**, shown in the figures as $f(x) = \|x\|$, where x is the error signal vector.

The notation for the optimization is $\hat{q} = \underset{q}{\operatorname{argmin}} \{f(x, q)\}$, indicating in words:

“find the best q that minimizes $f(q)$ ”.

Impulse Response (IR) Estimation Because the IR is used here with the intent to characterize the transducer and transmitter, the effects of the medium itself are not considered. Note that the problem of obtaining the actual values of the requisite IR data vector is beyond the scope of this document.

5. Types of Use Models

When the user defines a desired reference signal, he/she has two choices of where this signal “lives”: either in the domain of the acoustic medium, or in the domain of (electrical) received signal. Respectively, these are denoted the one-way and two-way problems.

The user also has the choice of whether the transducer is allowed to modify the desired reference waveform specification by filtering, or is to reproduce the desired reference waveform as closely as possible without filtering it. Respectively, these are denoted the **synthesis** and **compensation** problems. The term “synthesis” (or DAC synthesis) implies an emphasis on creating “analog” signals from a tri-state source. The term “compensation” refers to mitigating the non-ideal passband and transition-band characteristics of the transducer’s impulse response. Note that in either the synthesis case or the compensation case, the actual tri-state transmit sequence will be filtered by the transducer and transmit pulser hardware.

Therefore, the domain of desired reference signal needs to be specified. This choice constitutes the **Use Model**. The choice of use models described here classifies as one of the four cases below:

- * One-way synthesis: specify what the analog electrical signal is to be, before conversion to acoustic form (don’t care what transducer filtering will occur).
- * Two-way synthesis: specify what the received signal is to be, before it is filtered by transducer. Effects of the transducer will not be mitigated.

* One-way compensation: specify what the acoustic signal is to be (transmission will match the design signal, in the acoustic medium).

* Two-way compensation: specify what the received signal is to be (transmission will match the design signal, at the receiver).

The following figures (Figs. 1 - 4) depict these four cases. Note in the above cases, it is understood that bulk delay and receiver anti-aliasing filtering are implied. The acoustic medium is assumed to be transparent, and its effects are not otherwise considered by the encoding algorithm.

Figure 1. One-way Synthesis (DAC Synthesis) Use Model

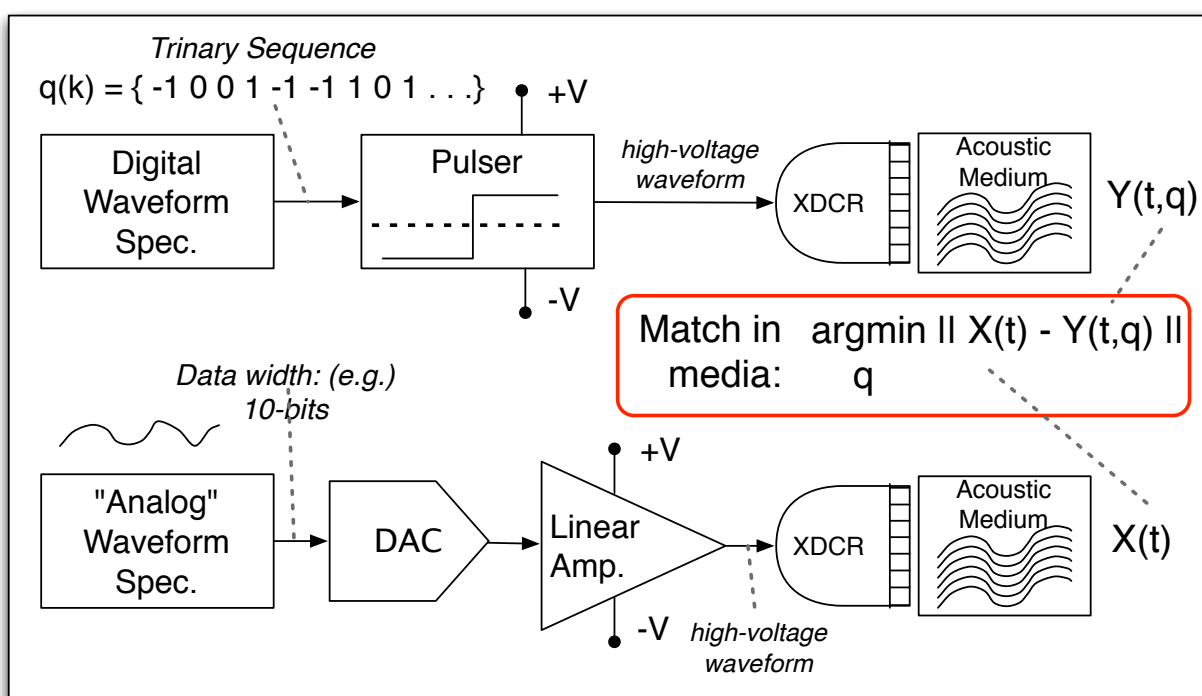


Figure 2. Two-way Synthesis (DAC Synthesis) Use Model

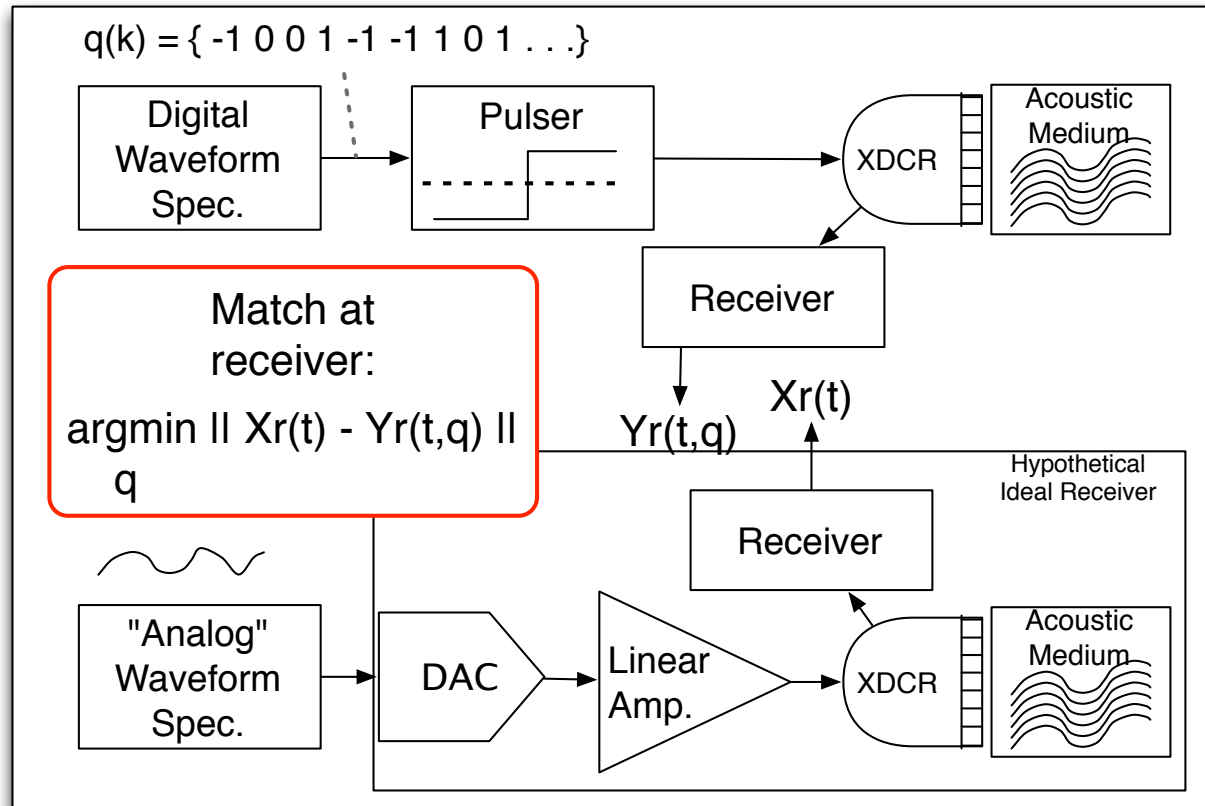
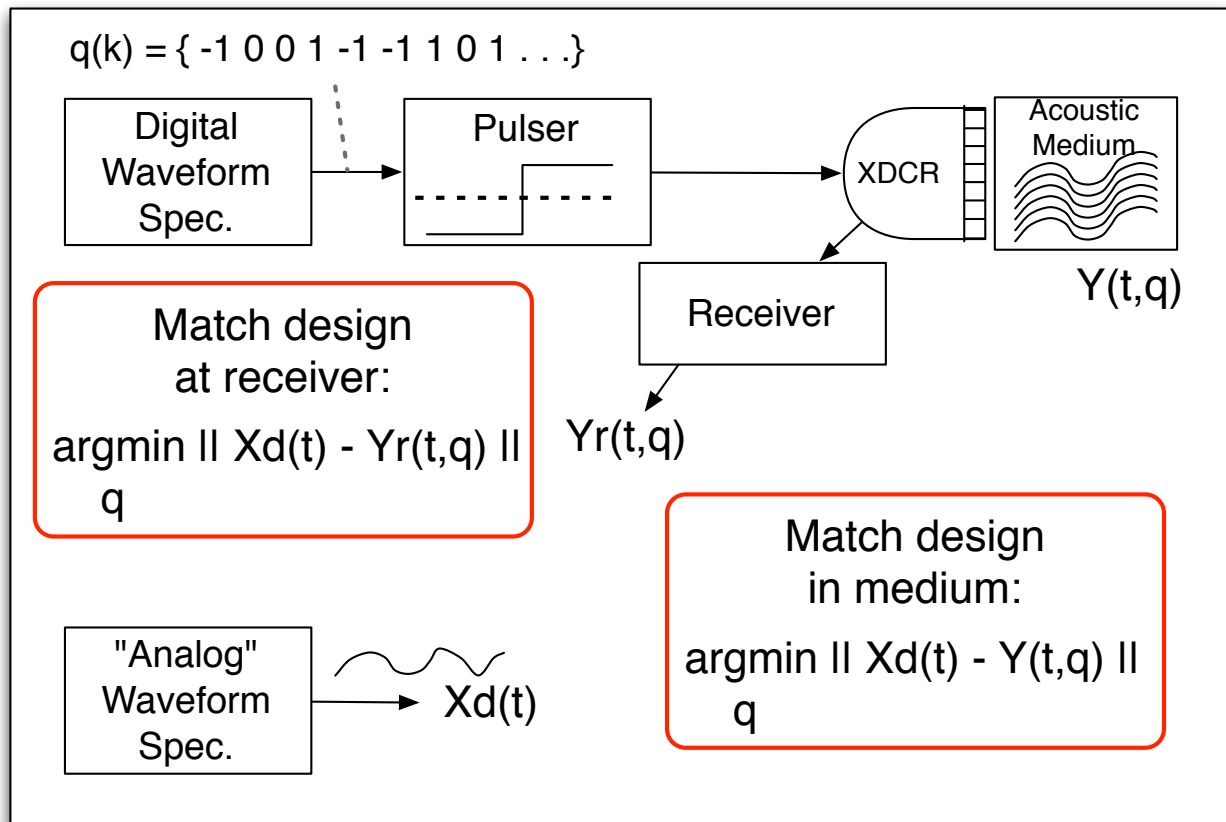


Figure 3. Compensation Use Models: One-Way (match in medium) and Two-Way (match at receiver)



6. The Two Encoder Implementations

The user has a choice of using two AW encoder tools. These include the GUI-based encoder (version 1), and the Command Line-based encoder (version 2). The encoders each entail a different user experience in terms of interface and execution time, as well as different performance capabilities.

The version 1 encoder has a MatLab GUI interface and runs relatively quickly. The version 2 encoder uses a command-line or programmatic scripting interface in the MatLab programming language, and has higher encoding accuracy than the version 1 encoder, but takes longer to execute for the same problem specification. For more details on specific encoder usage, see their individual user manuals (listed in the Bibliography) as appropriate.

7. Conclusion

While this document describes Use Models and explains their relation to encoders, the user is left to choose the Use Model and encoder most appropriate for the application and desired workflow.

8. Helpful References / Companion Documents

For background on the Arbitrary Waveform (**AW**) encoding problem in context of the Verasonics system, including usage models, please refer to the following documents:

New Command-Line Arbitrary Waveform Encoder (CLAW) version v2 (this document):

file: ArbWaveEncoder_v2_user_manual.pdf

Flynn, John A. "Command Line Arbitrary Waveform Encoder (CLAW) Version 2.0 User Manual"

Previous Arbitrary Waveform Generation Toolbox GUI tool (GUI-AWE) version v1:

file: ArbWaveToolbox_v1_Documentation.pdf

Verasonics staff, "Arbitrary Waveform Generation Toolbox"

USPTO Patent application for AW encoding methods including Viterbi:

file: ArbWave-patent-app-WO2015009960.pdf

Flynn, John A., et al. "Method and system for arbitrary waveform generation using a tri-state transmit pulser." U.S. Patent Application Nos. 14/905,434.

Presentation explaining linear AW encoding and usage models:

file: IEEE-2013_Verasonics-arbwave-presentation.pdf

Flynn, John A., et al. "Arbitrary waveforms using a tri-state transmit pulser." *Ultrasonics Symposium (IUS), 2013 IEEE International, Session IUS1-A3-4 Medical Signal Processing* (presentation materials).

Conference paper on linear AW encoding methods:

<http://ieeexplore.ieee.org/document/6725161/>

Flynn, John A., et al. "Arbitrary waveforms using a tri-state transmit pulser." *Ultrasonics Symposium (IUS), 2013 IEEE International* (pp 41-44). IEEE, 2013.